

Zeros of Polynomial Functions (2.3)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $\{-3, 1\}$

_____ Find the zeros of $f(x) = x^2 - x - 6$.

ANSWER: $(x - 1)(x + 4)$

_____ State the multiplicity of $x = 0$ in $f(x) = x^3(x - 2)$.

ANSWER: yes

_____ Find the zeros of $f(x) = x(x - 5)(x + 1)$.

ANSWER: no

_____ Write the factor corresponding to a zero at $x = -2$.

ANSWER: $\{-2, 3\}$ # _____ Is $x = 2$ a zero of $f(x) = x^2 - 4$?	ANSWER: $\{-4, 4\}$ # _____ Is $x = 1$ a zero of $f(x) = x^2 - 4$?
ANSWER: $(1, 2)$ # _____ Write the factors of a polynomial with zeros 1 and -4.	ANSWER: $(x + 2)$ # _____ Find the zeros of $f(x) = (x - 1)^2(x + 3)$.
ANSWER: $\{-1, 0, 5\}$ # _____ By the IVT, if $f(1) = -2$ and $f(2) = 3$, a zero lies in which interval?	ANSWER: 3 # _____ Find the zeros of $f(x) = x^2 - 16$.